

SUPPORTING INFORMATION

Significant Quantum Effects in Hydrogen Activation

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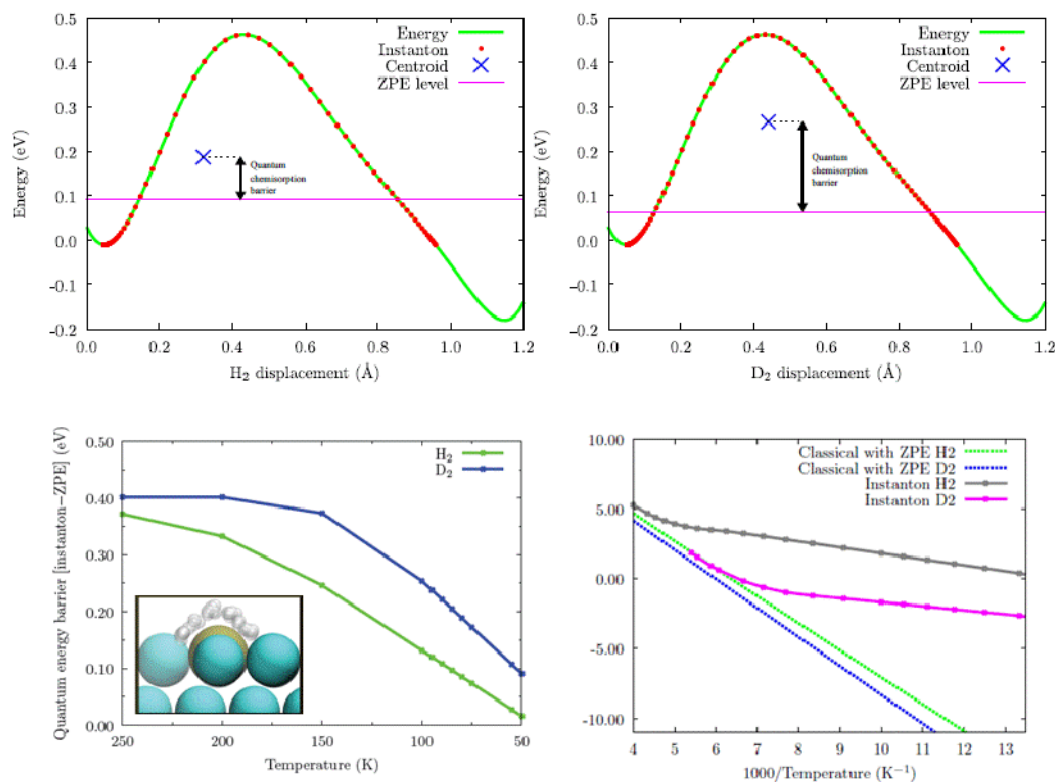


Figure S1. The top graphs show the spread of HQTST beads (red dots) at 85 K over the one-dimensional barrier that was computed with DFT (green line) for both hydrogen and deuterium dissociative chemisorption (top left and top right respectively). Spreading of the beads across the barrier (going from the precursor physisorbed state to the chemisorbed state) is a signature for quantum tunneling through the barrier. A clear isotope effect is visible as the centroid heights (shown as blue crosses) are at different heights for hydrogen and deuterium, and the zero-point energy levels being at different heights (pink line), resulting in a higher effective quantum barrier for deuterium. The graph bottom left shows how the effective quantum barrier (accounting for both zero-point energy and tunneling) changes with temperature, with the inset snapshot showing the spread of beads from path-integral molecular dynamics at 85 K. The plot on the bottom right shows the quantum effects lead to a marked change in the rate of the chemisorption process when compared with the classical rates.

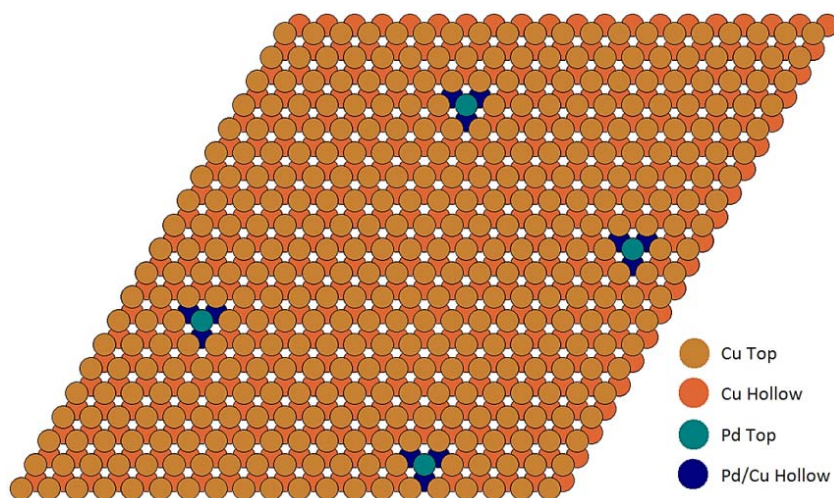


Figure S2. Sample lattice for KMC simulations of H_2/D_2 uptake. KMC simulations were performed on a 20x20 periodic Cu(111) grid with 1% random surface substitutions of Pd atoms. Four types of sites were treated explicitly in the model, as shown in the figure above. H_2/D_2 were allowed to occupy only Cu and Pd top sites, while H/D were allowed to occupy only Cu and Pd hollow sites.

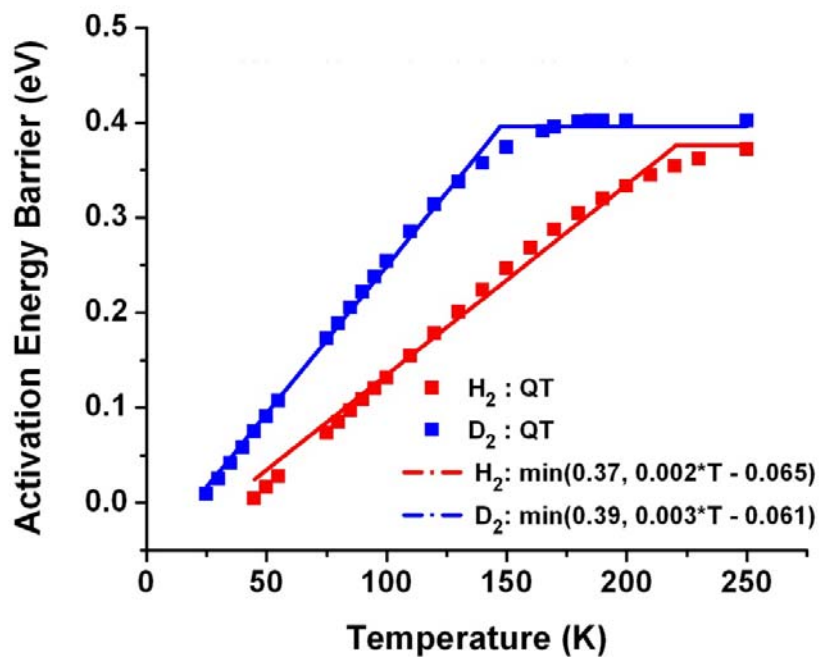


Figure S3. Activation energy barriers for the H_2/D_2 dissociation step over Pd surface sites. The symbols represent the quantum tunneling (QT) barriers obtained using the quantum path-integral molecular dynamics simulations and the solid lines denote the linear fits used for that process in our kMC calculations.

Summary of Adjustments to DFT parameters in the KMC simulations

KMC simulations performed using the DFT derived parameters (binding energies, activation energy barriers etc.) as the initial guesses to the model yielded no quantifiable H₂/D₂ uptake over the entire temperature range. Systematic adjustments in several of these DFT-derived parameters were needed to obtain a reasonable fit to experimentally observed trends. For the H₂ uptake simulations, molecularly adsorbed H₂ was stabilized on Pd-atop surface sites, and H adatoms were stabilized on all hollow sites by 0.10 eV. The slope and intercept of the linear function describing the temperature-dependent barrier for H₂ dissociation over Pd sites were adjusted in a way that resulted in a marginally higher barrier in the low temperature region and a slightly lower barrier in the high-temperature region when compared to the initial guesses, which were derived from the instanton calculations, as shown in Figure S4(a). Similarly, for the D₂ uptake simulations, adsorbed D₂ was stabilized on Pd-atop surface sites by 0.20 eV, and D adatoms were stabilized on all hollow sites by 0.15 eV. The intercept of the temperature-dependent activation energy barrier for D₂ dissociation was increased by 0.07 eV, and the classical barrier was decreased by 0.09 eV to fit the experimental trend, as shown in Figure S4(b). A summary of the adjustments made to the DFT-derived parameters is shown in Table S1.

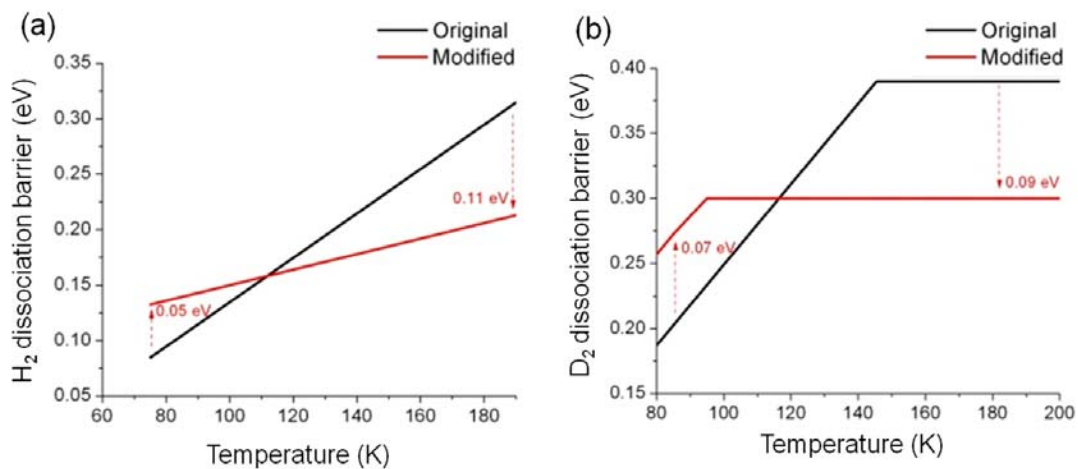


Figure S4. Adjustments to the kinetic barriers for (a) H₂ dissociation and (b) D₂ dissociation on the Pd/Cu sites, as required in our KMC simulations to obtain good fits with the experimental trends. The quantum tunneling barrier was lower than the classical barrier for H₂ dissociation over the entire temperature region of interest. The transition from quantum tunneling barrier to classical barrier is evident in the D₂ case, as the (zero slope) classical barrier becomes energetically preferred at higher temperatures.

Table S1. Activation energy barriers for elementary steps included in KMC simulations. All energy barriers for H₂ and D₂ processes are the same, with the exception of H₂/D₂ dissociation (Step 3) over surface Pd sites, as noted below.

Elementary Step	E _{act_forward_DFT} (eV)	E _{act_forward_adjusted} (eV)
(1) H ₂ (g) ↔ H ₂ (phy,Pd)	0.00	0.00
(2) H ₂ (g) ↔ H ₂ (phy,Cu)	0.00	0.00
(3) H ₂ (phy,Pd) ↔ 2H (PdCu)	min(0.37 ^a , 0.002*T – 0.0654 ^b)	min(0.37, 0.0007*T + 0.08)
D ₂ (phy,Pd) ↔ 2D (PdCu)	min(0.39 ^a , 0.0031*T – 0.061 ^b)	min(0.30, 0.0031*T + 0.009)
(4) H ₂ (phy,Cu) ↔ 2H (Cu)	0.74 ^a	0.74
(5) H ₂ (phy,Pd) ↔ H ₂ (phy,Cu)	0.12	0.27 ^c
(6) H ₂ (phy,Cu) ↔ H ₂ (phy,Cu)	0.12	0.27 ^c
(7) H (PdCu) ↔ H (Cu)	0.12	0.27 ^c
(8) H (Pd) ↔ H (Pd)	0.12	0.27 ^c
(9) H (Cu) ↔ H (Cu)	0.12	0.27 ^c

^a ZPE corrected activation energy barriers obtained from DFT calculations

^b Linear functions used to describe the temperature-dependent quantum tunneling barriers obtained from quantum path-integral calculations

^c The forward and reverse rate constants for diffusion events were reduced by several orders of magnitude (corresponding to a higher activation energy) while maintaining a constant K_{eq} to avoid spending prohibitively large amounts of time simulating these events, thereby reducing the computational requirements for KMC simulations. All modifications in these rate constants were verified for convergence at each temperature by comparing them to the simulations with full diffusion rates to ensure that the results were unaffected by the reduction in rate constants.

‘g’ and ‘phy’ in the parenthesis refer to gas-phase and physisorbed species, respectively.